Unit 2 Practice Test: Systems of Matter and Energy

Earth & Space Science

Name: _____ Period: _____ Date: _____

How to Work Through a Regents Question Set
1. Find the topic. Read the title and the first sentence of the reading.
2. Read the visuals first. Read every title, key, label, axis and unit on each diagram, graph or table before you start the questions.
3. Mark the reading. Underline numbers, processes and key words you might need later.
4. Circle the command word. *Identify* = name it. *Describe* = tell what happens. *Explain* = tell why or how. *Refute* = prove it wrong.
5. Answer with evidence. Point to the exact model, number or *Reference Tables* page you used. Each set starts with a **Skill** box. Use its strategy.

Base your answer to question 1 on the information below and on your knowledge of Earth and Space Sciences. Some questions may require the use of the *Reference Tables*.

Heat in Yellowstone

At Yellowstone National Park, magma several kilometers below the surface heats the solid rock above it. Groundwater that seeps down through cracks is heated by the hot rock. The heated water rises back toward the surface while cooler water sinks to replace it, and some of the water erupts from geysers like Old Faithful. During the day, the Sun also warms the ground at the surface.

Skill: Classify how heat is transferred.
Strategy: Decide for each process separately. Touching = conduction. A moving liquid or gas = convection. Energy traveling as waves, even through space = radiation.

1 Which row correctly identifies the type of heat transfer in each process?

Row	Hot rock heats the groundwater touching it	Heated water rises and cooler water sinks	Sun warms the ground at the surface
(1)	Conduction	Convection	Radiation
(2)	Convection	Conduction	Radiation
(3)	Radiation	Convection	Conduction
(4)	Conduction	Radiation	Convection

Base your answer to question 2 on the information below and on your knowledge of Earth and Space Sciences. Some questions may require the use of the *Reference Tables*.

Rockfalls in Yosemite Valley

More than a thousand rockfalls have been recorded in Yosemite Valley. One way a rockfall can begin is when water seeps into cracks in the granite cliffs. On cold winter nights, the water in the cracks can freeze. During the day, it thaws again. The table below shows the density of water in two phases.

Phase of Water	Density (g/cm^3)
Liquid water	1.00
Ice	0.92

Skill: Use data to explain a cause and effect.
Strategy: Compare the two densities. A lower density means the same mass takes up more space. Explain what that does inside a crack, then why repeating it every day and night matters.

2 Use the data in the table to explain how repeated freezing and thawing of water can widen the cracks in Yosemite's granite cliffs. [1]

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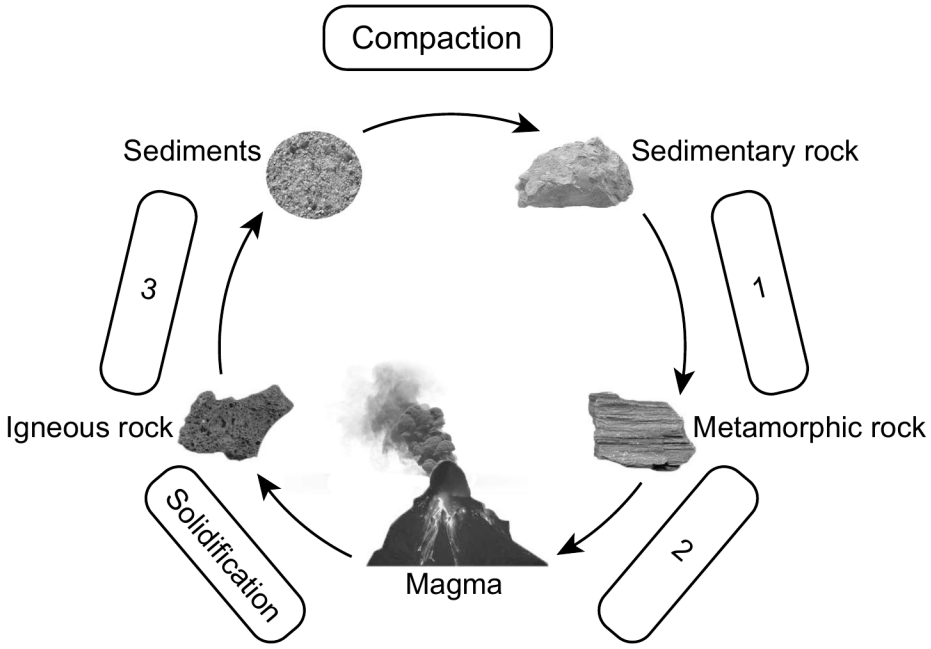
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Base your answers to questions 3 through 6 on the information below and on your knowledge of Earth and Space Sciences. Some questions may require the use of the *Reference Tables*.

Death Valley

Death Valley National Park, located on the border of California (CA) and Nevada (NV), is the largest national park in the continental U.S. It has the lowest elevation in North America at 282 feet below sea level and contains the driest desert in North America. Rock strata in the Death Valley region can provide evidence for the geologic history of the area. The oldest rocks are metamorphic and are approximately 1.7 billion years old. Sedimentary rocks in the valley region are about 500 million years old and are evidence that the area was the site of a warm shallow sea. Igneous rock layers are believed to range from 65.5 to about 2 million years old when volcanic ash and cinder from nearby volcanoes covered the region.

The rock cycle model below shows the formation of rocks as a result of sedimentary, igneous, and metamorphic processes.



Skill: Use a model to identify processes.
Strategy: For each numbered arrow, look at the rock it starts from and the rock it points to. Ask: what has to happen to change the first into the second?

3 Which list identifies the names of the missing processes, numbered 1 through 3, of the rock cycle?

(1) 1 = erosion; 2 = deformation; 3 = melting

(2) 1 = deformation; 2 = melting; 3 = erosion

(3) 1 = melting; 2 = deformation; 3 = erosion

(4) 1 = deformation; 2 = erosion; 3 = melting

Skill: Identify the energy sources that drive a system.
Strategy: Sort the processes in the model: which happen at Earth's surface (weathering, erosion) and which happen deep underground (melting, metamorphism)? Each group has its own energy source.

4 Which two sources of energy are responsible for the cycling of Earth materials and the flow of energy, as shown in the rock cycle model?

(1) the Sun and Earth's hot interior

(2) Earth's hot interior and the Moon's hot interior

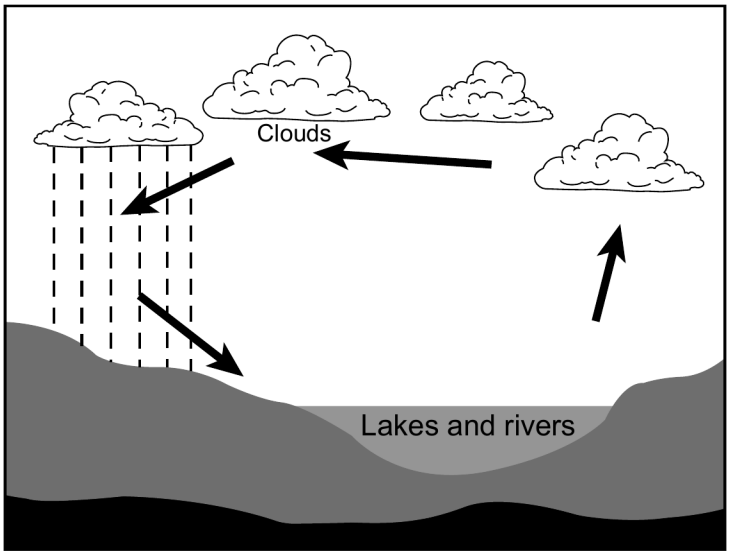
(3) evaporation and condensation of water on Earth's surface

(4) the Sun's energy and Earth's magnetic fields

The salt flats region in Death Valley covers over 200 square miles. Salt flat formation requires:

- a large basin where streams carry salts down to the basin
- an enclosed basin, like Death Valley, where the salts dissolved in stream water can be trapped and not wash away
- an arid climate with high temperatures to drive water cycle processes

The arrows on the water cycle model below represent the ways water changes its state as it moves through multiple pathways.



Skill: Explain how energy drives a process.
Strategy: Use the bullets in the reading to find the climate clue. Name the process, then describe what the Sun's energy does to the water molecules and what is left behind.

5 Identify **one** water cycle process that is directly responsible for the formation of the large mineral salt flats in Death Valley and explain how this process is driven by energy from the Sun. [1]

Process: _____

Explanation: _____

_____

A student collected a sample of the white crust from the salt flats at Badwater Basin and recorded these observations:

- Colorless to white with a vitreous (glassy) luster
- Breaks into cubes along three cleavage directions at 90°
- Does not scratch glass, and is scratched by a copper penny
- Does not bubble when acid is placed on it
- Tastes salty

Skill: Select the evidence that matters.
Strategy: Follow the *Mineral Identification Flowchart* (pp. 16–17) with the observations. Then find calcite on the flowchart. Which observation would be different for calcite?

6 Based on the observations and the *Mineral Identification Flowchart*, which choice identifies the mineral and the observation that best separates it from calcite?

(1) halite; it breaks along three cleavage directions at 90°

(2) halite; it has a vitreous luster

(3) calcite; it does not scratch glass

(4) gypsum; it is scratched by a copper penny

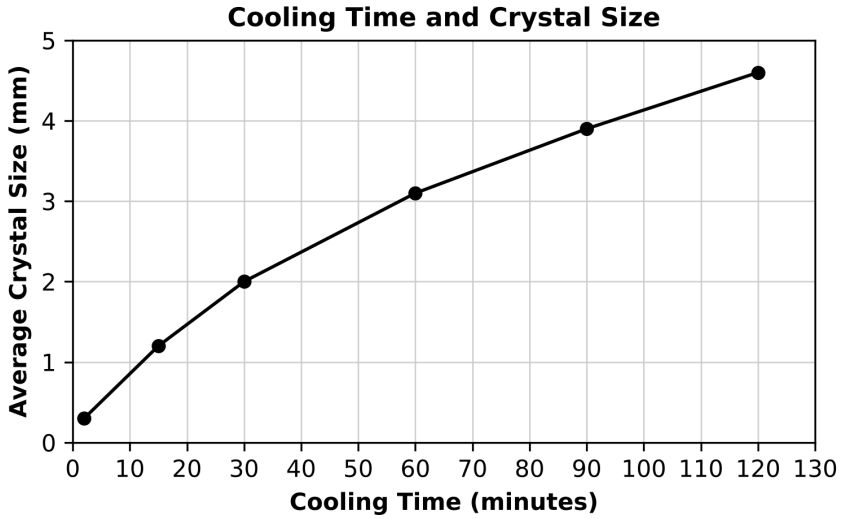
Base your answers to questions 7 through 10 on the information below and on your knowledge of Earth and Space Sciences. Some questions may require the use of the *Reference Tables*.

Igneous Rocks of Two National Parks

Yosemite National Park and Hawai‘i Volcanoes National Park both contain igneous rock, but the rocks formed in very different places. A student compared a rock sample from each park.

Sample	Location	Color	Average Crystal Size	Main Minerals
A	El Capitan, Yosemite	Light	5 mm	Quartz, potassium feldspar, sodium-rich plagioclase
B	Lava flow, Kilauea, Hawai‘i	Dark	Less than 1 mm	Pyroxene, calcium-rich plagioclase, olivine

The student also did an investigation. Melted samples of the same crystal-forming material were cooled for different amounts of time, and the average crystal size was measured. The graph shows the results.



Skill: Use the Reference Tables to classify.
Strategy: Find each sample's minerals on the *Mineral Composition of Igneous Rocks* chart (p. 14) to decide felsic or mafic. Then use crystal size to choose the rock name.

7 Which row correctly identifies each sample?

Row	Sample A	Sample B
(1)	Basalt, mafic	Granite, felsic
(2)	Granite, felsic	Basalt, mafic
(3)	Granite, mafic	Basalt, felsic
(4)	Gabbro, mafic	Rhyolite, felsic

Skill: Use data from a graph as evidence.
Strategy: Read the title and both axes first. Describe the pattern ("As cooling time increases…"), then use it to explain the samples. Name where each one cooled.

8 Use evidence from the graph to explain why Sample A has larger crystals than Sample B. In your answer, describe where each sample most likely cooled. [1]

Volcanic glass, such as obsidian, forms when lava cools so quickly that crystals do not have time to form. Obsidian is shiny, breaks with sharp curved edges, and is made mostly of silica.

Skill: Apply a definition.
Strategy: Check the reading against each part of the definition of a mineral: naturally occurring, inorganic, solid, definite chemical composition, and an orderly crystal structure. Which part is missing?

9 Which statement best explains why obsidian is **not** classified as a mineral?

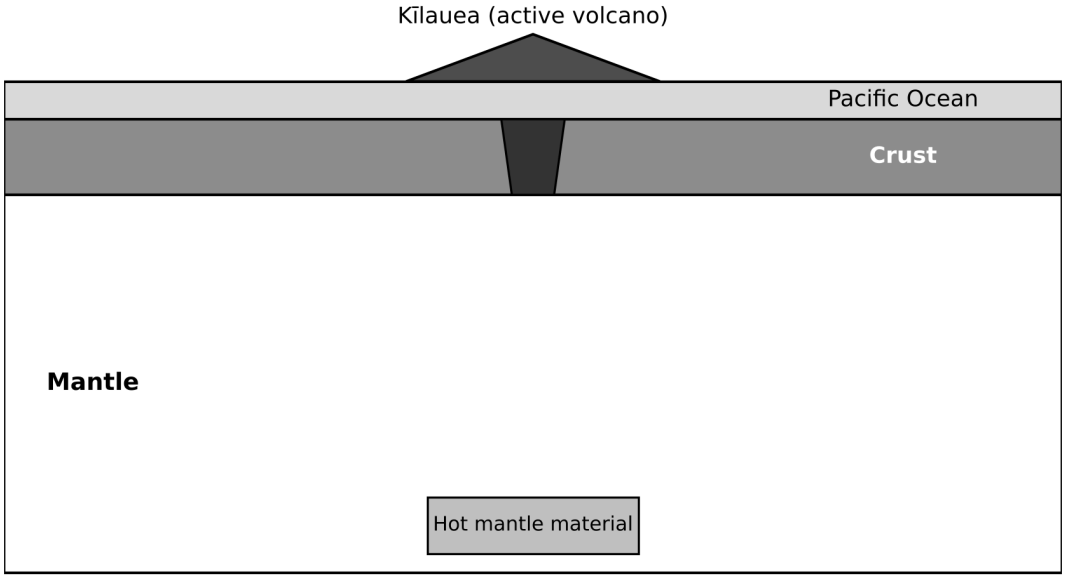
(1) It does not form naturally.

(2) It is made by living things.

(3) It does not have an orderly crystal structure.

(4) It is not a solid at Earth's surface.

The cross section below shows the mantle beneath Hawai‘i Volcanoes National Park. Hot mantle material deep below the island supplies the magma for Kilauea.



(Not drawn to scale)

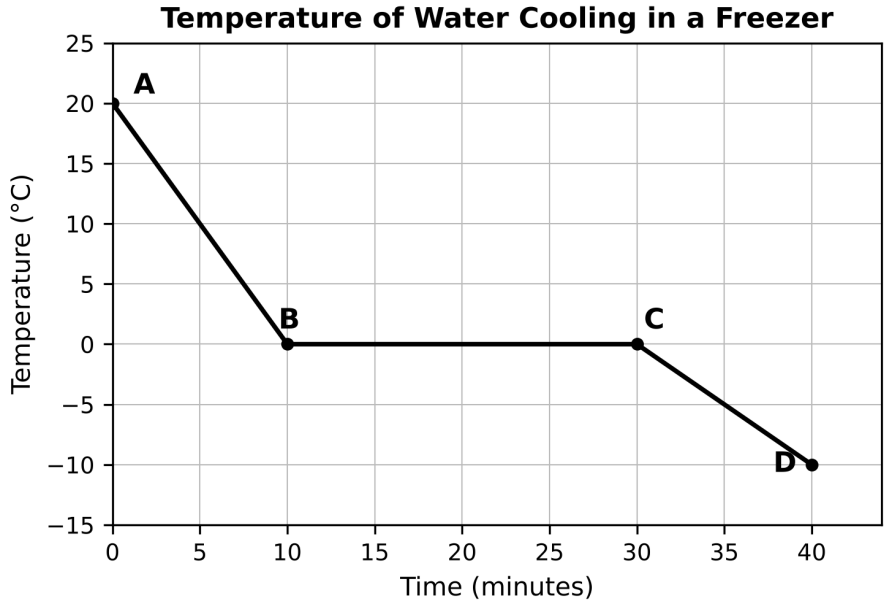
Skill: Develop a model.
Strategy: Hot material is less dense, so it rises. Start your arrows at the hot mantle material and show the direction it moves. Make sure each arrow clearly points one way.

10 On the cross section above, draw at least **two** arrows in the mantle to show how convection carries hot mantle material up toward Kilauea. [1]

Base your answers to questions 11 through 12 on the information below and on your knowledge of Earth and Space Sciences. Some questions may require the use of the *Reference Tables*.

Freezing Water

Ponds and lakes in Western New York often freeze over in winter. To model what happens as a pond freezes, a class placed a cup of water at 20°C into a freezer set at –15°C. The students recorded the temperature of the water every minute. The freezer removed energy from the water the entire time. The graph below shows the results. Points *A*, *B*, *C*, and *D* are labeled on the graph.



Skill: Read a graph to describe a phase change.
Strategy: Find the interval on the graph and read the temperature at each end. A flat line means the temperature is not changing. Ask what phase change happens at 0°C as water loses energy.

11 Which row in the table below correctly describes what happens to the water between points *B* and *C*?

Row	Phase Change	Energy	Temperature
(1)	Freezing	Absorbed by the water	Decreases
(2)	Condensation	Released by the water	Stays the same
(3)	Freezing	Released by the water	Stays the same
(4)	Melting	Absorbed by the water	Stays the same

Skill: Explain a pattern in data.
Strategy: Point to the exact part of the graph. Your answer must say where the energy leaving the water is going if it is not lowering the temperature.

12 The freezer removed energy from the water during the entire investigation. Use evidence from the graph to explain why the temperature of the water did **not** change between points *B* and *C*. [1]

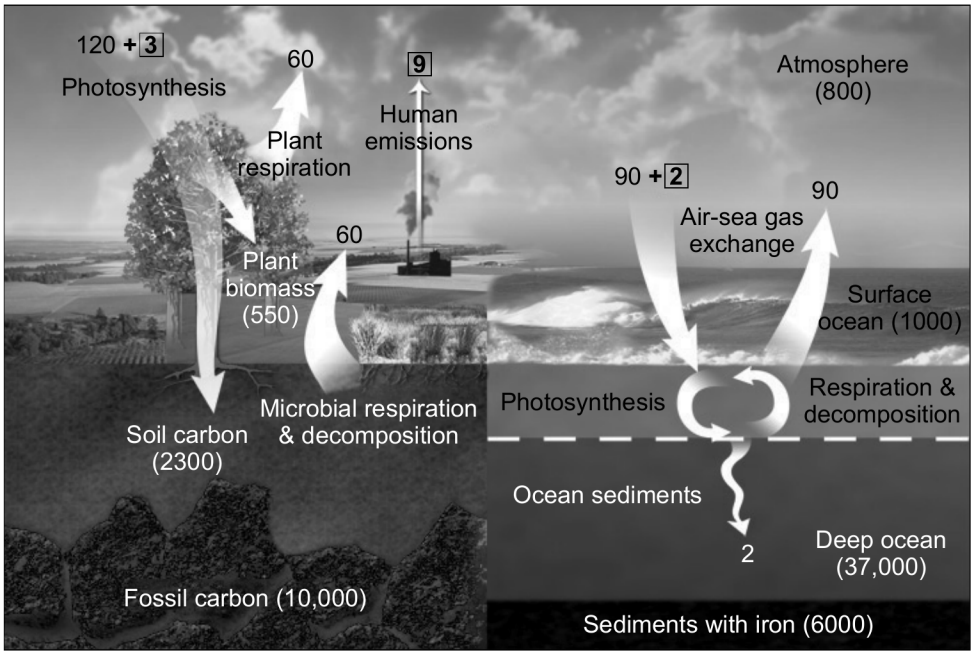
Base your answer to question 13 on the information below and on your knowledge of Earth and Space Sciences. Some questions may require the use of the *Reference Tables*.

The Carbon Cycle

The global carbon cycle refers to the movement of the element carbon through different storage places, or reservoirs, on Earth. Carbon moves through these reservoirs at different rates. Most carbon near Earth's surface cycles fairly quickly. Carbon in the atmosphere recycles in about three to five years, while plants recycle carbon in about 50 years. The carbon found in soil and fossil reservoirs is recycled, on average, in about 3000 to 5000 years.

The model below shows the movement of carbon (arrows) through the four Earth spheres. The numbers indicate the amount of carbon naturally added or removed from the spheres in gigatons (GT) per year. The numbers in bold indicate the amount of carbon added or removed by human activity. Numbers in parentheses ( ) are amounts of stored carbon.

Model of Fast Carbon Cycle



Skill: Use numbers from a model to compare.
Strategy: Test each choice against the model. Find every number the choice uses, check that it is attached to the right arrow or reservoir, and check the arrow's direction.

13 Which statement correctly identifies the quantitative cycling of carbon between two of Earth's spheres as a result of natural processes and human activities?

(1) Fossil carbon releases 2300 GT, while microbial respiration and decomposition absorb 60 GT.

(2) The deep ocean stores 36,000 more GT of carbon than is released by air-sea gas exchange.

(3) Human emissions add nine times more carbon to the atmosphere than plant respiration, which is the same amount released to the atmosphere by microbial respiration and decomposition.

(4) The amount of carbon that leaves the atmosphere and is absorbed by the ocean is 92 GT, which is the same amount that is released by the oceans back into the atmosphere and absorbed by ocean sediments.